

C++ Basic PROGRAMS

ATUL KUMAR (LINKEDIN).
NOTES GALLERY (TELEGRAM).

1]. FIBONACCI SERIES IN C++



```
#include <iostream>
using namespace std;
int main ( )
{
    int n1, n2, n3, i, number;
    n1 = 0;
    n2 = 1;
    Cout << "enter the number of elements: ";
    Cin >> number;
    Cout << n1 << n2 << " " : || Printing 0 & 1
    for (i = 2; i < number; i++)
    {
        n3 = n1 + n2
        cout << n3 << " ";
        n1 = n2;
        n2 = n3;
    }
    return 0;
}
```

Output:- enter the number of element : 10
0 1 1 2 3 5 8 13 21 34.

2]. PRIME NUMBER PROGRAM IN C++



```
#include <iostream>
using namespace std;
int main ( )
```

SWIPE ➡


```
→ {  
    int n, i, m = 0, Flag = 0;  
    Cout << "enter number to check prime : ";  
    Cin >> n;  
    m = n / 2;  
    for (i = 2; i <= m; i++)  
    {  
        if (n % i == 0)  
        {  
            Cout << "number is not prime" << endl;  
            Flag = 1;  
            break;  
        }  
    }  
    if (Flag == 0) {  
        Cout << "number is prime" << endl;  
    }  
    return 0;  
}
```

Output:-

- enter a number to check prime : 20
number is not prime.
- enter a number to check prime : 17
number is prime.

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3]. PALINDROME NUMBER IN C++

→ The palindrome number is same after reverse.
Example:- 121, 32123, 2372732

```
→ #include <iostream>  
using namespace std;  
int main ( )
```


→

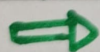
```
{
    int n, r, sum = 0, temp;
    cout << "enter the number = ";
    cin >> n;
    temp = n;
    while (n > 0)
    {
        r = n % 10;
        sum = (sum * 10) + r;
        n = n / 10;
    }
    if (temp == sum)
        cout << "number is palindrome.";
    else
        cout << "number is not palindrome.";
    return 0;
}
```

Output :- • enter the number = 12321
number is Palindrome.

• enter the number = 120
number is not Palindrome.

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4]. FACTORIAL PROGRAM IN C++



```
#include <iostream>
using namespace std;
int main ( )
{
    int i, fact = 1, number;
    cout << "enter any number : ";
    cin >> number;
```




```
for (i = 1; i <= number; i++) {  
    fact = fact * i;  
}  
cout << "factorial of " << number << " is :"  
      << fact << endl;  
return 0;  
}
```

Output:- enter any number : 5
factorial of 5 is : 120.

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5]. C++ PROGRAM TO SWAP TWO NUMBER WITHOUT USING THIRD VARIABLE :

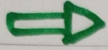


```
#include <iostream>  
using namespace std;  
int main ()  
{  
    int a = 20 , b = 50.  
    cout << "Before swap a = " << a << " b = " << b;  
    a = a + b || a = 70 ( 20 + 50 )  
    b = a - b || b = 20 ( 70 - 50 )  
    a = a - b || a = 50 ( 70 - 20 )  
    cout << "after swap a = " << a << " b = " << b;  
    return 0;  
}
```

Output:-

Before swap a = 20 , b = 50.
After swap a = 50 , b = 20.

6].

C++ PROGRAM TO PRINT HALF STAR PYRAMID PATTERN:

```
#include <iostream>
using namespace std;
int main ( )
{
    int n, j, i;
    cout << "enter the number of rows : ";
    cin >> n;
    for (i = 1, i <= n, i++)
    {
        for (j = 1, j <= i, j++)
        {
            cout << " * ";
        }
        cout << "\n ";
    }
    return 0;
}
```

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enter number rows : 5

```
★
★ ★
★ ★ ★
★ ★ ★ ★
★ ★ ★ ★ ★
```


7].

C++ PROGRAM FOR FULL ★ PYRAMID :

```
#include <iostream>
using namespace std;
int main ( )
{
    int rows i j spaces ;
    cout << "enter number rows : " ;
    cin >> rows ;
    for ( i = 1 , i <= rows ; i ++ )
    {
        for ( spaces = i ; spaces < rows ; spaces ++ )
        {
            cout << "   " ;
        }
        for ( j = 1 ; j <= 2 * i - 1 ; j ++ )
        {
            cout << " ★ "
        }
        cout << " \n " ;
    }
    return 0 ;
}
```

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```
★
★ ★
★ ★ ★
★ ★ ★ ★
★ ★ ★ ★ ★
★ ★ ★ ★ ★ ★
★ ★ ★ ★ ★ ★ ★
```

enter number of rows
: 7

8].

**SUM OF DIGIT PROGRAM C++ :**

#include <iostream>

using namespace std;

int main()

{

int n, sum = 0, m;

cout << "enter a number :";

cin >> n;

while (n > 0)

{

m = n % 10

sum = sum + m;

n = n / 10;

}

cout << "sum is = " << sum << endl;

return 0;

}

Output:-

enter 0 number = 55

sum is 10.

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